Noncommutative Cumulant Frontiers for the Fabius–Rvachev Law

Free and Boolean probability, exact Hankel obstructions, geometric sinc products, Jacobi stripping, and dyadic acceleration

Research report prepared from the `ProveIt/Analysis/FabiusFunction/docs` corpus

30 August 2026

Abstract

The Fabius function and Rvachev's compactly supported C^∞ function up sit at the meeting point of dyadic functional equations, the Thue–Morse sequence, infinite sinc products, Bernoulli and Bell polynomials, q-series, orthogonal polynomials, and Lambert–W endpoint asymptotics. The accompanying repository corpus develops these classical, analytic, arithmetic, and representation-theoretic directions in great depth. This report adds a different layer: free and Boolean cumulants of the probability law with density up, together with their interaction with the existing Jacobi, q-binomial, finite-sinc, inverse-Fabius, Legendre, and endpoint frameworks.

The main paper-level proved result is an exact rational certificate that the Rvachev up-law is *not* freely infinitely divisible. After variance normalization, a 3×3 shifted free-cumulant Hankel block has negative determinant; an even simpler polynomial test vector gives a negative quadratic form. We extend this obstruction to the geometric-uniform family

$$Y_q = \sum_{k \geq 1} q^k U_k, \qquad U_k \overset{\text{iid}}{\sim} \mathrm{Unif}[-1, 1],$$

proving non-free-infinite-divisibility for every $0 < q < 0.523498599534498\ldots$, including $q = 1/2$. The endpoint is the square root of a rigorously isolated root of an explicit degree-thirty integer polynomial.

Further exact bridges are obtained. Lagrange inversion turns the Bernoulli–Bell moment formula into a finite formula for free cumulants. Boolean cumulants are identified with moments of the once-stripped Jacobi measure, proving strict positivity. Every fixed-order free or Boolean cumulant of a finite dyadic sinc product is an exact polynomial in 4^{-N}, so the repository's $q = 1/4$ Richardson and q-binomial acceleration machinery applies directly. A Legendre-coefficient transform and an inverse-Fabius integral transform are given explicitly. Finally, symbolic evidence through r_{16}, exact positivity through r_{120}, and endpoint computations motivate conjectures on Jacobi increments, free-cumulant positivity, and a Lambert–W-controlled transform radius.

Status convention. Statements labelled theorem, proposition, lemma, or corollary are proved in this paper. Statements labelled example record exact or high-precision calculations reproducible by the included Python program. Conjectures are explicitly marked.

Arrival provenance, inherited overlap, and Lean boundary. This package arrived on 30 August 2026 as `drafts/incoming/Fabius_Rvachev_Noncommutative_Frontiers.zip`, introduced at repository commit `536c5b42d71e9048b2399f5929fd8fdf7cd34c04`. Its exact outer SHA-256 is `55f780d0780a693f2450fe6a4c8a63ba964b3d0e6fcea6d985040c6cb29e25cc`; the original archive bytes remain recoverable from that commit.

The dyadic probability law, classical moment and cumulant recurrences, Bernoulli–Bell formulas, Jacobi and continued-fraction representations, finite sinc products, q-binomial and Richardson machinery, inverse-Fabius and Legendre transforms, and Lambert–W endpoint analysis are inherited interfaces from the repository corpus. This report's distinct focus is the free- and Boolean-cumulant transform layer, its exact non-free-infinite-divisibility certificates, and the associated conjectures; it does not claim the inherited interfaces anew.

Paper theorem labels are not Lean declarations. At intake, the live Lean tree had no exact counterpart for the free or Boolean cumulants, the shifted free-cumulant Hankel obstruction, the parameter-uniform obstruction, or the Jacobi-stripping results stated here. Those claims remain research-frontier formalization obligations; only separately cross-referenced inherited inputs may be treated as Lean-backed.

Contents

1 Corpus audit, novelty screen, and scope

1.1 The audited documentation tree

The requested source tree is not a single article but a layered research corpus. Its live top level contains a mathematical glossary, the principal Fabius–Rvachev exposition, a non-elementarity study, a Lean walkthrough, an archive, a paper collection, and both semi-formalized and non-formalized frontier branches. The semi-formalized branch further records a canonical consolidated source, a proof-dependency manifest, a gap register, and many thematic drafts.

The frontier manifest is especially important for a novelty audit. It states that eleven earlier notebooks were consolidated into six main syntheses and a gap register, and it records provenance for work on dyadic formulae, q-connections, small-argument and endpoint asymptotics, repeated integration, inverse-function saddles, and Thue–Morse convergence. It also lists later large volumes devoted to representations, exponents and q-series, polynomial synthesis, dyadic combs and finite sinc products, all-orders inverse endpoint asymptotics, Lambert–W, and a broad q-Pochhammer/q-binomial monograph.

Consequently, several apparently natural directions are not new relative to the repository. A report centered only on Jacobi coefficients, Legendre expansions, Mellin transforms, Padé approximation, finite sinc products, Richardson extrapolation, or Lambert–W endpoint inversion would duplicate substantial existing material. The present report uses those results as interfaces and develops a noncommutative cumulant layer not found in the audited sources.

1.2 New contributions

The main additions are:

N1. an exact free-cumulant transform of the Bernoulli–Bell moment formulas;

N2. an exact proof that the dyadic up-law is not freely infinitely divisible;

N3. a parameter-uniform obstruction for $0 < q < 0.5234985995\ldots$;

N4. a Boolean-cumulant coefficient-stripping theorem for the Jacobi continued fraction;

N5. exact 4^{-N}-polynomial laws for free and Boolean cumulants of finite dyadic sinc products;

N6. inverse-Fabius and finite Legendre transforms for these cumulants;

N7. a Jacobi-increment positivity conjecture verified through r_{16};

N8. endpoint conjectures connecting Lambert–W flatness to transform radii.

The novelty claim is relative to the repository corpus. Free and Boolean cumulants and their general relations to Jacobi parameters are established topics; the new content is their specialization to the Fabius–Rvachev law, the exact rational and q-parametric certificates, and the bridges to the existing dyadic machinery.

1.3 Notation and normalization

Write $U \sim \mathrm{Unif}[-1, 1]$, so

$$\mathbb{E}e^{tU} = \frac{\sinh t}{t}, \qquad \mathbb{E}e^{itU} = \operatorname{sinc} t := \frac{\sin t}{t}.$$

Let $U_1, U_2, \ldots$ be independent copies. For $0 < q < 1$, define

$$Y_q := \sum_{k=1}^{\infty} q^k U_k. \tag{1}$$

Its support is $[-q/(1-q), q/(1-q)]$. At $q = 1/2$, the support is $[-1, 1]$, and the density is Rvachev's up function normalized by $\int_{-1}^{1} \mathrm{up}(x)\, dx = 1$. The law for negative q is the same as for $|q|$, because the summands are symmetric.

The repository also uses a unit-support geometric normalization with weights $(1-q)q^k$. It differs only by a nonzero scalar. Free infinite divisibility and variance-normalized cumulants are invariant under scaling, so (1) is algebraically convenient.

2 Classical dyadic structure as probabilistic input

2.1 Infinite sinc products and the Rvachev law

Independence gives

$$\phi_q(t) := \mathbb{E}e^{itY_q} = \prod_{k=1}^{\infty} \mathrm{sinc}(q^k t), \tag{2}$$

$$\mathcal{M}_q(t) := \mathbb{E}e^{tY_q} = \prod_{k=1}^{\infty} \frac{\sinh(q^k t)}{q^k t}. \tag{3}$$

For $q = 1/2$, (2) is the standard infinite sinc product for up. The random series gives the fixed point

$$Y_{1/2} \overset{d}{=} \frac{Y'_{1/2} + U}{2}, \tag{4}$$

where $Y'_{1/2}$ is an independent copy.

2.2 Bernoulli cumulants

Use the convention $B_2 = 1/6$, $B_4 = -1/30$. Since

$$\log \frac{\sinh t}{t} = \sum_{m\geq 1} \frac{2^{2m} B_{2m}}{2m} \frac{t^{2m}}{(2m)!},$$

classical cumulants add and scale to give the following.

Proposition 2.1 (Classical cumulants). *For $m \geq 1$,*

$$\kappa_{2m}(Y_q) = \frac{2^{2m} B_{2m}}{2m} \frac{q^{2m}}{1-q^{2m}}, \qquad \kappa_{2m+1}(Y_q) = 0. \tag{5}$$

At $q = 1/2$,

$$\kappa_{2m} = \frac{4^m B_{2m}}{2m(4^m - 1)}. \tag{6}$$

Proof. Apply the uniform cumulant expansion to each $q^k U_k$, then sum $\sum_{k\geq 1} q^{2mk} = q^{2m}/(1 - q^{2m})$. □

Let $m_n = \mathbb{E}Y^n$. The complete exponential Bell polynomial gives

$$m_n = B_n(\kappa_1, \ldots, \kappa_n), \tag{7}$$

and equivalently

$$m_n = \sum_{j=1}^{n} \binom{n-1}{j-1} \kappa_j m_{n-j}, \qquad m_0 = 1. \tag{8}$$

2.3 A positive dyadic moment recurrence

Proposition 2.2 (Stable dyadic moment recurrence)**.** *For* $m_{2n} = \mathbb{E}Y_{1/2}^{2n}$,

$$(4^n - 1)m_{2n} = \sum_{k=1}^{n} \binom{2n}{2k} \frac{m_{2n-2k}}{2k+1}, \qquad m_0 = 1. \tag{9}$$

Proof. Raise (4) to power $2n$, expand, and use symmetry. The $k = 0$ term is $4^{-n}m_{2n}$, and $\mathbb{E}U^{2k} = 1/(2k+1)$. □

3 Free and Boolean transforms of the moment sequence

3.1 Partition cumulants

Let

$$M(z) = 1 + \sum_{n\geq 1} m_n z^n.$$

The free cumulants r_n are defined by

$$m_n = \sum_{\pi\in \mathrm{NC}(n)} \prod_{V\in\pi} r_{|V|}. \tag{10}$$

Equivalently, if $C(w) = 1 + \sum_{n\geq 1} r_n w^n$, then

$$M(z) = C(zM(z)). \tag{11}$$

The Boolean cumulants b_n, associated with interval partitions, satisfy

$$M(z) = \frac{1}{1 - B(z)}, \qquad B(z) = \sum_{n\geq 1} b_n z^n. \tag{12}$$

For a symmetric law, write

$$A(t) := \sum_{n\geq 0} m_{2n} t^n, \quad D(w) := 1 + \sum_{n\geq 1} r_{2n} w^n, \quad H(t) := \sum_{n\geq 1} b_{2n} t^n. \tag{13}$$

Then

$$A(t) = D\big(tA(t)^2\big), \qquad H(t) = 1 - \frac{1}{A(t)}. \tag{14}$$

3.2 Lagrange inversion

Theorem 3.1 (Lagrange formula for free cumulants)**.** *For any moment sequence with* $m_0 = 1$,

$$r_n = -\frac{1}{n-1}[z^n]M(z)^{1-n}, \qquad n \geq 2. \tag{15}$$

For a symmetric sequence,

$$r_{2n} = -\frac{1}{2n-1}[t^n]A(t)^{1-2n}. \tag{16}$$

Proof. Set $w = zM(z)$, and let $z = z(w)$ be its compositional inverse. By (11), $C(w) = M(z(w))$. Lagrange–Bürmann inversion gives

$$[w^n]C(w) = \frac{1}{n}[z^{n-1}]M'(z)M(z)^{-n}.$$

Since $(M^{1-n})' = (1-n)M^{-n}M'$, coefficient extraction yields (15). Symmetry gives $M(z) = A(z^2)$, hence (16). □

Expanding $(1+X)^{1-2n}$ gives a completely finite formula.

Corollary 3.2 (Finite composition formula). *For $n \geq 1$,*

$$r_{2n} = \frac{1}{2n-1} \sum_{k=1}^{n} (-1)^{k+1} \binom{2n+k-2}{k} \sum_{\substack{j_1+\cdots+j_k=n \\ j_i \geq 1}} \prod_{i=1}^{k} m_{2j_i}. \tag{17}$$

The Boolean analogue is

$$b_{2n} = \sum_{k=1}^{n} (-1)^{k+1} \sum_{\substack{j_1+\cdots+j_k=n \\ j_i \geq 1}} \prod_{i=1}^{k} m_{2j_i}. \tag{18}$$

Combining (5), (7), and (17) is an explicit Bernoulli–Bell–noncrossing transform. It converts the classical Bernoulli cumulants of the infinite sinc product into free cumulants without integration or numerical inversion.

3.3 Triangular recurrences

From (14),

$$A(t) = 1 + \sum_{j \geq 1} r_{2j} t^j A(t)^{2j}, \tag{19}$$

$$b_{2n} = m_{2n} - \sum_{j=1}^{n-1} b_{2j} m_{2n-2j}. \tag{20}$$

Thus

$$r_{2n} = m_{2n} - \sum_{j=1}^{n-1} r_{2j} [t^{n-j}] A(t)^{2j}. \tag{21}$$

These recurrences are the basis of the exact rational computations in the accompanying program.

3.4 First exact dyadic values

Table 1: First even moments, free cumulants, and Boolean cumulants of the dyadic up-law.

order	m_{2n}	r_{2n}	b_{2n}
2	1/9	1/9	1/9
4	19/675	7/2025	32/2025
6	583/59535	562/893025	4384/893025
8	132809/32531625	55603/379535625	6845728/3415820625
10	46840699/24405225075	148564406/3294705385125	109557322976/115314688479375
12	4068990560161/4133856862760625	3386539245383834/2148572104419834843 75	1066748816562523 84/214857210441983484375

Example 3.3 (Exact positivity check). The program computed r_{2n} by rational arithmetic for $1 \leq n \leq 60$, and every value was positive. Thus positivity is verified exactly through r_{120}, not merely numerically.

4 An exact obstruction to free infinite divisibility

4.1 Conditional positive definiteness

For a compactly supported probability measure, free infinite divisibility implies conditional positive definiteness of its free-cumulant sequence. In one variable, a necessary condition is

$$\sum_{i,j \geq 1} \overline{a_i} a_j r_{i+j} \geq 0 \tag{22}$$

for every finitely supported sequence $(a_i)_{i\geq 1}$. Equivalently, every finite principal or nonprincipal submatrix of the shifted Hankel matrix

$$\mathcal{H}_r = (r_{i+j})_{i,j\geq 1}$$

must be positive semidefinite. For a symmetric law, odd free cumulants vanish, so this matrix separates into odd- and even-index parity blocks.

4.2 The dyadic certificate

Theorem 4.1 (The Rvachev up-law is not freely infinitely divisible). *Let $Y = Y_{1/2}$, and let $X = 3Y$, so $\mathrm{Var}(X) = 1$. Write*

$$c_{2n} := r_{2n}(X) = 3^{2n} r_{2n}(Y).$$

Then the shifted even block

$$H = \begin{pmatrix} c_4 & c_6 & c_8 \\ c_6 & c_8 & c_{10} \\ c_8 & c_{10} & c_{12} \end{pmatrix} \tag{23}$$

has negative determinant. Hence neither X nor Y is freely infinitely divisible.

Proof. The exact normalized free cumulants are

$$c_2 = 1, \qquad c_4 = \frac{7}{25}, \qquad c_6 = \frac{562}{1225},$$
$$c_8 = \frac{500427}{520625}, \qquad c_{10} = \frac{148564406}{55796125}, \qquad c_{12} = \frac{3386539245383834}{404291747234375}.$$

Direct rational elimination gives

$$\det H = -\frac{110020031172003013633803653}{3288379747918347944189453125} < 0. \tag{24}$$

Therefore H is not positive semidefinite, contradicting the necessary condition (22). Scaling by a nonzero constant preserves free infinite divisibility, so the conclusion holds for Y as well. $\square$

A shorter certificate avoids a determinant.

Corollary 4.2 (Polynomial witness). *For $v = (3, -4, 1)^\top$,*

$$v^\top H v = -\frac{108624693817191}{404291747234375} < 0. \tag{25}$$

Equivalently, the conditional cumulant functional is negative on

$$p(x) = 3x^2 - 4x^4 + x^6 = x^2(x^2 - 1)(x^2 - 3).$$

Remark 4.3 (Positivity is not enough). The first sixty even free cumulants are positive, yet the shifted Hankel form is indefinite. This cleanly separates coefficientwise positivity from the much stronger conditional positive definiteness required by free infinite divisibility.

4.3 A three-way divisibility comparison

The up-law is nondegenerate and compactly supported, so it is not classically infinitely divisible: a nondegenerate classically infinitely divisible law cannot have bounded support. Theorem 4.1 shows it is not freely infinitely divisible either. In contrast, every probability measure is Boolean infinitely divisible, because Boolean convolution powers are defined by scalar multiplication of the Boolean cumulant series. Thus the up-law gives a concrete trichotomy:

classical ID: no, free ID: no, Boolean ID: yes.

5 A geometric q-family obstruction

5.1 Variance-normalized free cumulants

Put $s = q^2$ and normalize Y_q to variance one. Since

$$\kappa_2(Y_q) = \frac{s}{3(1-s)},$$

define

$$c_{2n}(s) := \frac{r_{2n}(Y_q)}{r_2(Y_q)^n}.$$

These quantities depend only on s, as expected from symmetry and scale invariance.

Proposition 5.1 (First normalized free cumulants). *For* $0 < s < 1$,

$$c_2(s) = 1, \tag{26}$$

$$c_4(s) = \frac{11s-1}{5(1+s)}, \tag{27}$$

$$c_6(s) = \frac{2(379s^3 + 20s^2 + 20s + 1)}{35(1+s)(1+s+s^2)}, \tag{28}$$

$$c_8(s) = \frac{3P_8(s)}{175(1+s)^2(1+s^2)(1+s+s^2)}, \tag{29}$$

$$P_8(s) = 24359s^6 + 4221s^5 + 4579s^4 + 4242s^3 + 379s^2 + 21s - 1.$$

The formulas for c_{10} *and* c_{12} *are given in Appendix A.*

Proof. Insert (5) into the Bell recurrence (8), apply the triangular free recurrence (21), and divide by r_2^n. Every operation is rational in s, so the displayed identities reduce to polynomial verification. □

The first obstruction is immediate: $c_4(s) < 0$ for $0 < s < 1/11$. For larger s, a higher shifted-Hankel block is required.

5.2 The degree-thirty determinant

Define

$$D_3(s) := \det\begin{pmatrix} c_4(s) & c_6(s) & c_8(s) \\ c_6(s) & c_8(s) & c_{10}(s) \\ c_8(s) & c_{10}(s) & c_{12}(s) \end{pmatrix}. \tag{30}$$

Exact simplification gives

$$D_3(s) = \frac{P_{30}(s)}{11802415625(1+s)^6(1+s^2)^3(1-s+s^2)(1+s+s^2)^4(1+s+s^2+s^3+s^4)^2}, \tag{31}$$

where $P_{30} \in \mathbb{Z}[s]$ is listed in Appendix B. The denominator is positive on $(0,1)$.

Let α be the largest root of P_{30} in $(0,1)$. Exact Sturm-sequence computations isolate it in

$$\frac{385546279}{1406842461} < \alpha < \frac{219810848}{802080713} \tag{32}$$

and give

$$\alpha = 0.2740507837145811031976360832564018082588497489 68\ldots. \tag{33}$$

The same Sturm certificate shows that there are no roots of P_{30} in $(1/11, \alpha)$. Since $D_3(1/4) < 0$, it follows that $D_3(s) < 0$ throughout that interval.

Theorem 5.2 (Geometric non-free-infinite-divisibility interval)**.** *Let Y_q be defined by* (1)*. If*

$$0 < q < \sqrt{\alpha} = 0.52349859953449837591\ldots, \tag{34}$$

then Y_q is not freely infinitely divisible.

Proof. If $0 < s < 1/11$, then $c_4(s) < 0$, violating conditional positive definiteness. If $1/11 \le s < \alpha$, then $D_3(s) < 0$, so the block in (30) is not positive semidefinite. Since $s = q^2$, the stated range follows. □

Remark 5.3. The dyadic point $q = 1/2$ has $s = 1/4 < \alpha$, so Theorem 4.1 is recovered. The direct rational certificate of Section 4 is nevertheless preferable at the dyadic point because it is shorter and does not invoke root isolation.

5.3 A hierarchy of necessary-condition thresholds

For each $m \ge 1$, form the leading $m \times m$ even shifted-Hankel block

$$H_m(s) = \big(c_{2(i+j)}(s)\big)_{1\le i,j\le m}.$$

The following are the largest zeros detected numerically for $\det H_m(s)$ in the searched interval.

Table 2: Numerical leading-block thresholds. These are necessary-condition diagnostics, not a proof of free infinite divisibility above the listed values.

m	largest detected s-zero	$q = \sqrt{s}$
1	0.090909090909090909	0.301511344577764
2	0.199485938552759446	0.446638487540829
3	0.274050783714581103	0.523498599534498
4	0.320508711922987676	0.566134888452379
5	0.349023432785722262	0.590782051847991
6	0.366350957768133855	0.605269326637435
7	0.376359680090712126	0.613481605340137
8	0.381275825505797916	0.617475364290591
9	0.382775958248985252	0.618688902639271
10	0.382808454300472788	0.618715164110653

Remark 5.4 (Open problem: free-divisibility phase diagram). Determine the set

$$\mathcal{F} := \{q \in (0,1) : Y_q \text{ is freely infinitely divisible}\}.$$

The present work proves $\mathcal{F} \cap (0, \sqrt{\alpha}) = \varnothing$. Two plausible scenarios remain: a genuine threshold after which all laws are freely infinitely divisible, or failure for every finite $q < 1$ with free infinite divisibility appearing only in the Gaussian limit.

5.4 The Gaussian endpoint and connected partitions

Let $Z_q = Y_q/\sqrt{\operatorname{Var}(Y_q)}$. From (5), the standardized classical cumulants satisfy

$$\lambda_{2m}(s) = \frac{3^m 2^{2m} B_{2m}}{2m} \frac{(1-s)^m}{1-s^m}, \tag{35}$$

so $\lambda_2 = 1$ and $\lambda_{2m} = O((1-s)^{m-1})$ for $m \ge 2$. Thus Z_q converges in moments and distribution to the standard normal law as $q \uparrow 1$.

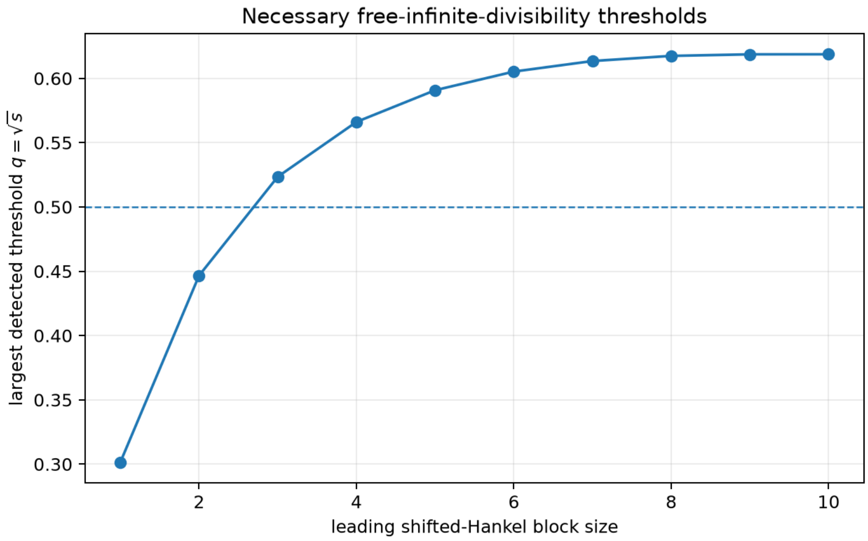

Figure 1: Largest detected zero of the leading even shifted-Hankel determinant. The dashed line marks the dyadic parameter $q = 1/2$.

The free cumulants of the standard normal enumerate connected pairings. Consequently,

$$(c_2, c_4, c_6, c_8, c_{10}, c_{12}) \longrightarrow (1, 1, 4, 27, 248, 2830). \tag{36}$$

Writing $\varepsilon = 1 - s$, exact expansion of Proposition 5.1 and Appendix A gives

$$\begin{aligned} c_4 &= 1 - \frac{3}{5}\varepsilon + O(\varepsilon^2), \\ c_6 &= 4 - \frac{27}{5}\varepsilon + O(\varepsilon^2), \\ c_8 &= 27 - \frac{306}{5}\varepsilon + O(\varepsilon^2), \\ c_{10} &= 248 - 831\varepsilon + O(\varepsilon^2), \\ c_{12} &= 2830 - \frac{65178}{5}\varepsilon + O(\varepsilon^2). \end{aligned} \tag{37}$$

Only $\lambda_4 = -(3/5)\varepsilon + O(\varepsilon^2)$ contributes at first order. Lehner's connected-partition formula therefore gives an enumerative interpretation.

Proposition 5.5 (First Gaussian correction)**.** *Let A_n be the number of connected set partitions of $[2n]$, in Lehner's sense, with one block of size four and all remaining blocks of size two. Then*

$$c_{2n}(s) = \gamma_n - \frac{3}{5}A_n(1 - s) + O((1 - s)^2), \tag{38}$$

where γ_n is the number of connected pairings of $[2n]$. For $n = 2, \dots, 6$,

$$A_n = 1, 9, 102, 1385, 21726.$$

Proof. Free cumulants are sums over connected set partitions weighted by classical cumulants. At $s = 1$, only $\lambda_2 = 1$ survives, giving connected pairings. To first order in $1 - s$, exactly one block may carry $\lambda_4 = -(3/5)(1 - s) + O((1 - s)^2)$; all other blocks are pairs. Higher classical cumulants contribute only at second or higher order. □

6 Boolean cumulants and Jacobi coefficient stripping

6.1 The Stieltjes continued fraction

The repository develops the symmetric Jacobi matrix and the Stieltjes continued fraction of the even moment series. Write

$$A(t) = \cfrac{1}{1 - \cfrac{\beta_1 t}{1 - \cfrac{\beta_2 t}{1 - \cfrac{\beta_3 t}{\ddots}}}}, \qquad \beta_j > 0. \tag{39}$$

For the up-law, the first coefficients include

$$\beta_1 = \frac{1}{9}, \qquad \beta_2 = \frac{32}{225}. \tag{40}$$

The associated Jacobi operator has zero diagonal, because the measure is symmetric.

Theorem 6.1 (Boolean coefficient stripping). *Let A have the S-fraction (39), and let*

$$A^{(1)}(t) = \cfrac{1}{1 - \cfrac{\beta_2 t}{1 - \cfrac{\beta_3 t}{\ddots}}}$$

be the once-stripped tail. Then

$$H(t) = 1 - \frac{1}{A(t)} = \beta_1 t A^{(1)}(t). \tag{41}$$

Consequently,

$$b_{2n} = \beta_1 m_{n-1}^{(1)} > 0, \tag{42}$$

where $m_k^{(1)} = [t^k]A^{(1)}(t)$ is the kth moment of the first associated Jacobi measure.

Proof. Remove the first denominator in (39):

$$A(t) = \frac{1}{1 - \beta_1 t A^{(1)}(t)}.$$

Substitution into $H = 1 - A^{-1}$ proves (41). Since $\beta_j > 0$, the tail is a Stieltjes moment series with positive coefficients; the up-law has infinite support, so the coefficients are strictly positive. □

Remark 6.2. This identifies Boolean cumulants not merely as recursively defined rational numbers but as a shifted moment sequence of an associated orthogonality measure. It creates a direct bridge between noncommutative probability and the repository's Jacobi/Legendre representation program.

6.2 Jacobi increments and free-cumulant positivity

Let

$$\Delta_j := \beta_{j+1} - \beta_j. \tag{43}$$

The first free cumulants, re-expanded in β_1 and the increments, are

$$r_2 = \beta_1, \tag{44}$$
$$r_4 = \beta_1\Delta_1, \tag{45}$$
$$r_6 = \beta_1\bigl(2\Delta_1^2 + \beta_2\Delta_2\bigr), \tag{46}$$
$$\begin{aligned} r_8 = \beta_1\bigl(&\beta_1^2\Delta_3 + 6\beta_1\Delta_1\Delta_2 + 2\beta_1\Delta_1\Delta_3 + 2\beta_1\Delta_2^2 + \beta_1\Delta_2\Delta_3 \\ &+ 5\Delta_1^3 + 6\Delta_1^2\Delta_2 + \Delta_1^2\Delta_3 + 2\Delta_1\Delta_2^2 + \Delta_1\Delta_2\Delta_3\bigr). \end{aligned} \tag{47}$$

These formulas were obtained by expanding the weighted-Dyck-path moments of (39), applying (21), and substituting $\beta_j = \beta_1 + \Delta_1 + \cdots + \Delta_{j-1}$.

Example 6.3 (Jacobi-increment positivity through r_{16})**.** The exact symbolic expansion has nonnegative integer coefficients at every checked order. The monomial counts are

cumulant	r_4	r_6	r_8	r_{10}	r_{12}	r_{14}	r_{16}
monomials	1	3	10	35	120	410	1400

All coefficients were checked exactly as integers.

Conjecture 6.4 (Jacobi-increment positivity)**.** *For every $n \geq 1$, r_{2n}/β_1 is a polynomial with nonnegative integer coefficients in*

$$\beta_1, \Delta_1, \dots, \Delta_{n-1}.$$

Therefore any symmetric measure with nondecreasing S-fraction coefficients β_j has nonnegative even free cumulants.

A proof would likely require a sign-free reinterpretation of Lehner's lattice-path formula after coefficient stripping. The repository conjectures monotone behavior of the up-law's Jacobi coefficients toward the arcsine limit $1/4$; Conjecture 6.4 would convert such monotonicity into free-cumulant positivity. Theorem 4.1 shows that this still would not imply free infinite divisibility.

7 Interfaces with inverse Fabius and Legendre expansions

7.1 An inverse-Fabius moment identity

On $[0, 1]$, the standard relation between the two functions is

$$\mathrm{up}(x) = F(1 - x), \tag{48}$$

with up even. Therefore

$$m_{2n} = 2\int_0^1 x^{2n} F(1-x)\,dx = 2\int_0^1 (1-y)^{2n} F(y)\,dy. \tag{49}$$

Integration by parts and the substitution $u = F(y)$ give a useful inverse-function formula.

Proposition 7.1 (Inverse-Fabius representation)**.** *For $n \geq 0$,*

$$m_{2n} = \frac{2}{2n+1}\int_0^1 \bigl(1 - F^{-1}(u)\bigr)^{2n+1}\,du. \tag{50}$$

Proof. In the last integral of (49), integrate $(1-y)^{2n}$ and differentiate $F(y)$. The boundary terms vanish because $F(0) = 0$ and $(1-y)^{2n+1} = 0$ at $y = 1$. Thus

$$m_{2n} = \frac{2}{2n+1}\int_0^1 (1-y)^{2n+1} F'(y)\,dy.$$

Now set $u = F(y)$. □

Combining (50) with Theorem 3.1 gives an explicit inverse-Fabius/free-cumulant bridge:

$$r_{2n} = -\frac{1}{2n-1}[t^n]\left(1 + \sum_{j\geq 1} \frac{2t^j}{2j+1} \int_0^1 (1 - F^{-1}(u))^{2j+1}\, du\right)^{1-2n}. \tag{51}$$

The corresponding Boolean formula is obtained by replacing the outer power with $1 - (\cdot)^{-1}$. Thus every free or Boolean cumulant is a finite polynomial in integral moments of the inverse Fabius function.

7.2 A finite transform from Legendre coefficients

Suppose the even Legendre expansion is

$$\mathrm{up}(x) = \sum_{k\geq 0} a_{2k} P_{2k}(x), \tag{52}$$

with the repository's normalization of P_n. Orthogonality is not needed for the following triangular transform; only the standard integral

$$\int_{-1}^{1} x^{2n} P_{2k}(x)\, dx = \frac{2(2n)!}{(2n-2k)!!(2n+2k+1)!!}, \qquad 0 \leq k \leq n. \tag{53}$$

Proposition 7.2 (Legendre-to-cumulant transform)**.** *The moment m_{2n} depends only on $a_0, a_2, \ldots, a_{2n}$, namely*

$$m_{2n} = \sum_{k=0}^{n} \frac{2(2n)!}{(2n-2k)!!(2n+2k+1)!!} a_{2k}. \tag{54}$$

Consequently, r_{2n} and b_{2n} are finite polynomial functions of the first $n+1$ even Legendre coefficients.

Proof. Insert (52) into $m_{2n} = \int_{-1}^{1} x^{2n}\, \mathrm{up}(x)\, dx$. Terms with $k > n$ vanish, and (53) yields (54). Apply (17) and (18). □

This is useful computationally in both directions. High-quality Legendre coefficients produce free and Boolean cumulants without sampling up; conversely, exact cumulants supply nonlinear consistency checks on numerically computed Legendre expansions.

8 Finite dyadic sinc products and exact $q = 1/4$ acceleration

8.1 Partial random series

Define the finite dyadic approximation

$$Y^{(N)} := \sum_{k=1}^{N} 2^{-k} U_k, \qquad Q := 4^{-N}. \tag{55}$$

Its characteristic function is the finite sinc product

$$\phi_N(t) = \prod_{k=1}^{N} \mathrm{sinc}(t/2^k),$$

and its density is a compactly supported spline whose signed knot structure is governed by dyadic subset sums and the Thue–Morse parity pattern. The classical cumulants truncate geometrically:

$$\kappa_{2m}^{(N)} = \kappa_{2m}(1 - Q^m). \tag{56}$$

Theorem 8.1 (Polynomial transfer law)**.** *Fix $n \geq 1$. Each of the quantities*

$$m_{2n}^{(N)}, \qquad r_{2n}^{(N)}, \qquad b_{2n}^{(N)}$$

is an exact polynomial in $Q = 4^{-N}$ of degree at most n. Its constant term is the corresponding cumulant or moment of the infinite up-law.

Proof. Assign weighted degree m to κ_{2m}. Formula (56) is a polynomial in Q of degree m. Every monomial in the Bell formula for m_{2n} has total weighted degree n, so substitution yields a polynomial of degree at most n. Free and Boolean cumulants are finite polynomials in moments in which each monomial contributing to order $2n$ has total compressed degree n, by (17) and (18). The constant term is obtained by setting $Q = 0$, which restores the full infinite cumulants. □

The first exact free-cumulant polynomials are

$$r_2^{(N)} = \frac{1-Q}{9}, \tag{57}$$

$$r_4^{(N)} = \frac{(Q-1)(43Q-7)}{2025}, \tag{58}$$

$$r_6^{(N)} = -\frac{2(Q-1)(8219Q^2-3100Q+281)}{893025}, \tag{59}$$

$$r_8^{(N)} = \frac{(Q-1)(11727403Q^3-6416697Q^2+1095297Q-55603)}{379535625}. \tag{60}$$

The first Boolean polynomials are

$$b_2^{(N)} = \frac{1-Q}{9}, \tag{61}$$

$$b_4^{(N)} = \frac{4(Q-1)(17Q-8)}{2025}, \tag{62}$$

$$b_6^{(N)} = -\frac{4(Q-1)(6829Q^2-5225Q+1096)}{893025}. \tag{63}$$

The included computation records exact formulas through order twelve.

8.2 Richardson extraction and q-binomial weights

Let $R = 1/4$, so $Q_N = R^N$. If

$$F_N = F(Q_N) = a_0 + a_1 Q_N + \cdots + a_n Q_N^n,$$

then a_0 can be recovered from finitely many consecutive levels by Lagrange interpolation at zero.

Proposition 8.2 (Exact geometric Richardson formula)**.** *For $p \geq 0$, define*

$$w_{p,k} := \frac{(-1)^{p-k} R^{(p-k)(p-k+1)/2}}{(R;R)_k (R;R)_{p-k}}, \qquad 0 \leq k \leq p, \tag{64}$$

where $(R;R)_j = \prod_{m=1}^{j}(1-R^m)$. Then

$$\mathcal{R}_p F_N := \sum_{k=0}^{p} w_{p,k} F_{N+k} = a_0 + O(Q_N^{p+1}). \tag{65}$$

If F is a polynomial of degree at most p, the equality $\mathcal{R}_p F_N = a_0$ is exact.

Proof. The weights are the Lagrange basis polynomials evaluated at zero for the nodes $1, R, \dots, R^p$:

$$w_{p,k} = \prod_{\substack{0 \le j \le p \\ j \ne k}} \frac{-R^j}{R^k - R^j}.$$

Separating factors with $j < k$ and $j > k$ reduces this product to (64). Interpolation proves (65). $\square$

Thus every fixed-order free or Boolean cumulant of the up-law can be recovered *exactly* from a sufficiently deep finite-sinc hierarchy. This imports the repository's q-Pochhammer and Gaussian-binomial acceleration apparatus into free probability with no new analytic error estimate.

Corollary 8.3 (Exact finite-level recovery)**.** *For fixed n, applying $\mathcal{R}_n$ to $r_{2n}^{(N)}$ or $b_{2n}^{(N)}$ returns the infinite-level cumulant exactly, for every $N \ge 1$.*

8.3 A Thue–Morse research direction

The density of $Y^{(N)}$ admits alternating truncated-power formulas whose signs are Thue–Morse values. Theorem 8.1 shows that after integrating these complicated splines against cumulant test polynomials, the entire dependence on level collapses to a degree-n polynomial in 4^{-N}. A worthwhile combinatorial problem is to derive the coefficients of (58)–(60) directly from Prouhet cancellation, without passing through classical cumulants. Such a proof would identify precisely which Thue–Morse cancellations correspond to noncrossing rather than ordinary set partitions.

9 Endpoint flatness, Boolean renewal, and a free transform radius

9.1 The endpoint constant

Define

$$S := A(1) = \sum_{n \ge 0} m_{2n} = \int_{-1}^{1} \frac{\mathrm{up}(x)}{1 - x^2}\, dx. \tag{66}$$

The integral converges because up is flat at $x = \pm 1$. The positive recurrence (9) through $n = 200$ gives the truncated diagnostics

$$S_{200} := \sum_{n=0}^{200} m_{2n} = 1.157443071998936858827816325718399902147132272887 8\dots, \tag{67}$$

$$S_{200}^{-2} = 0.7464499676168194568444653281203431295286709304 19\dots. \tag{68}$$

Also

$$A'_{200}(1) := \sum_{n=1}^{200} n m_{2n} = 0.24115244290218716016587027418686550821455195266 48\dots. \tag{69}$$

These are finite sums, not certified decimal expansions of S or $A'(1)$; no numerical tail bound is claimed here.

9.2 A conditional defective-renewal theorem

Write $a_n = m_{2n}$, $C(t) = A(t) - 1$, and $\rho = C(1) = S - 1$. From $H = C/(1 + C)$,

$$H(t) = C(t) - C(t)^2 + C(t)^3 - \cdots \tag{70}$$

inside the disk of convergence and formally coefficientwise.

Proposition 9.1 (Boolean asymptotics under a subexponential hypothesis)**.** *Assume the positive sequence* $(a_n)_{n\geq 1}$ *is summable and convolution-subexponential in the sense that, for each fixed* $k \geq 1$,

$$[t^n]C(t)^k \sim k\rho^{k-1}a_n. \tag{71}$$

Then

$$\frac{b_{2n}}{m_{2n}} \longrightarrow \frac{1}{(1+\rho)^2} = S^{-2}. \tag{72}$$

Proof. Insert (71) into (70). The asymptotic multiplier is

$$\sum_{k\geq 1}(-1)^{k+1}k\rho^{k-1} = \frac{1}{(1+\rho)^2}.$$

A standard dominated-tail argument justifies truncating the sum under the subexponential hypothesis. □

Conjecture 9.2 (Boolean endpoint ratio)**.** *The dyadic moment sequence satisfies* (71)*, and therefore* (72) *holds.*

At $n = 200$, the exact/high-precision computation gives

$$\frac{b_{400}}{m_{400}} = 0.7316693282109417\ldots,$$

which is slowly increasing toward the conjectured limit S^{-2}; the order-200 proxy is $S_{200}^{-2} = 0.7464499967\ldots$. The slow convergence is consistent with the logarithmic and Lambert–W scales already present in endpoint asymptotics.

9.3 The free compositional map

The free relation in (14) is controlled by

$$\Psi(t) := tA(t)^2, \qquad A(t) = D(\Psi(t)). \tag{73}$$

At the positive endpoint the exact identities are

$$\Psi(1) = S^2, \tag{74}$$

$$\Psi'(1) = S^2 + 2SA'(1). \tag{75}$$

The corresponding order-200 diagnostics are

$$S_{200}^2 = 1.339674464918336133231397224 4517\ldots,$$

$$S_{200}^2 + 2S_{200}A'_{200}(1) = 1.897914913583847579412435088 1873\ldots.$$

Because $\Psi'(1) > 0$, the real inverse is regular up to the endpoint. The obstruction to analytic continuation is not an algebraic branch point but the flat, nonanalytic endpoint singularity inherited from up.

Conjecture 9.3 (Positive free cumulants and endpoint radius)**.** *For the dyadic up-law:*

(i) $r_{2n} > 0$ *for every* $n \geq 1$;

(ii) the radius of convergence of $D(w) = 1 + \sum_{n\geq 1} r_{2n}w^n$ *is* S^2;

(iii) consequently,

$$\lim_{n\to\infty} r_{2n}^{1/n} = S^{-2}. \tag{76}$$

Part (i) is verified exactly through r_{120}. Parts (ii)–(iii) require excluding nearer complex critical points or singularities of the inverse of Ψ. The positive endpoint value is a natural candidate because, under coefficientwise positivity, Pringsheim's theorem forces a dominant singularity on the positive real axis.

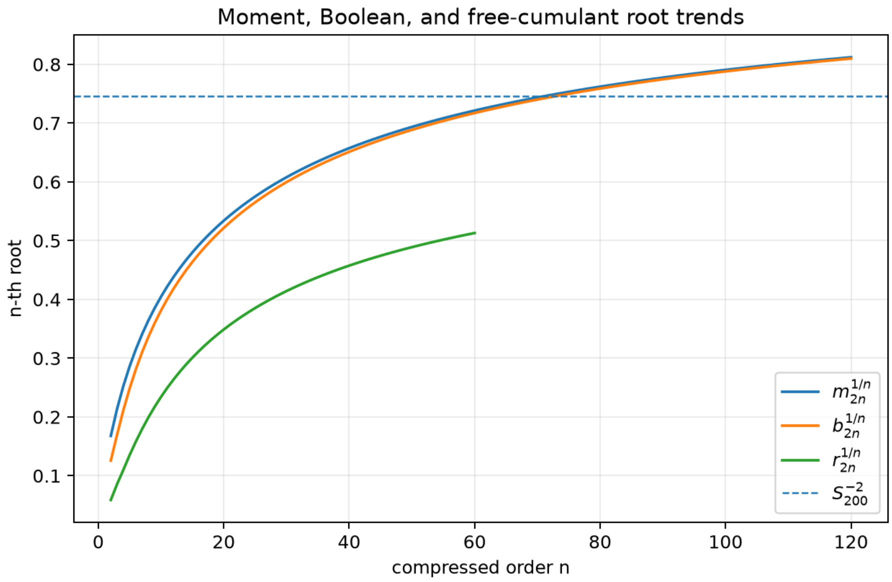

Figure 2: Truncated root diagnostics. The moment and Boolean roots tend slowly to one, while the free-cumulant roots appear to approach the plotted order-200 guide S_{200}^{-2}. Neither that guide nor the observed convergence is treated as a certified value or proof.

9.4 Why Lambert–W should reappear

The repository's endpoint analysis expresses the saddle location and the flat decay of $\mathrm{up}(1-x)$ through Lambert–W-type scales. Moments

$$m_{2n} = 2\int_0^1 x^{2n}\,\mathrm{up}(x)\,dx$$

are therefore controlled by a moving endpoint saddle whose distance from one is itself described by nested logarithms and Lambert–W. The Boolean transform $1 - A^{-1}$ preserves the endpoint location but changes its amplitude. The free transform additionally rescales the singular location by $\Psi(1) = S^2$. A full proof of Conjectures 9.2 and 9.3 should combine:

(a) the existing all-orders endpoint expansion for up;

(b) a uniform saddle-point estimate for m_{2n-k}/m_{2n} when k ranges from fixed to mesoscopic scales;

(c) a defective-renewal theorem for the Boolean coefficients;

(d) singular inversion of $\Psi(t) = tA(t)^2$ in a slit complex neighborhood of $t = 1$.

This is a concrete route by which Lambert–W enters noncommutative cumulant asymptotics rather than merely the original density.

10 Cyclotomic and q-integer structure

10.1 Normalized classical cumulants live in a q-integer ring

Let

$$[m]_s := \frac{1-s^m}{1-s} = 1 + s + \cdots + s^{m-1}.$$

After variance normalization, (35) can be written

$$\lambda_{2m}(s) = \frac{3^m 2^{2m} B_{2m}}{2m} \frac{(1-s)^{m-1}}{[m]_s}. \tag{77}$$

Thus every denominator is cyclotomic: its complex zeros are roots of unity other than one.

Proposition 10.1 (Cyclotomic denominator control)**.** *For fixed n, the variance-normalized moment, free cumulant $c_{2n}(s)$, and Boolean cumulant belong to*

$$\mathbb{Q}\left[s, [2]_s^{-1}, [3]_s^{-1}, \dots, [n]_s^{-1}\right]. \tag{78}$$

In particular, they have no pole on the positive real interval $0 < s < 1$.

Proof. The normalized classical cumulants of orders at most $2n$ belong to the ring (78) by (77). Moments, free cumulants, and Boolean cumulants of order $2n$ are universal polynomials with rational or integer coefficients in those classical cumulants. The ring is closed under these operations. Since $[m]_s > 0$ on $(0, 1)$, no positive pole occurs. □

The explicit denominators in Proposition 5.1 and Appendix A are cancellations and factorizations of products of $[m]_s$. For example,

$$[2]_s = 1 + s, \quad [3]_s = 1 + s + s^2, \quad [4]_s = (1 + s)(1 + s^2),$$

while $1 - s + s^2$ is the sixth cyclotomic polynomial and arises after cancellation among the $[m]_s$.

10.2 A q-binomial expansion problem

The ring statement is structural but not yet canonical. It suggests looking for formulas of the form

$$c_{2n}(s) = \sum_{\lambda \vdash n-1} \frac{P_{n,\lambda}(s)}{\prod_j [j]_s^{\lambda_j}}, \tag{79}$$

where $P_{n,\lambda}$ has a sign, positivity, or palindromicity property and the sum is indexed by connected partition types. Such a formula would merge three existing structures:

(i) Bernoulli numbers from the uniform summands;

(ii) noncrossing/connected partitions from free cumulants;

(iii) q-integers and Gaussian-binomial identities from the geometric weights.

Conjecture 10.2 (Cyclotomic numerator regularity)**.** *After multiplication by a minimal product of $[j]_s$, the numerator of $c_{2n}(s)$ has a finite decomposition into reciprocal or anti-reciprocal pieces indexed by connected partition block profiles. The decomposition, rather than the fully expanded numerator, has integer coefficients of controlled sign.*

The raw numerator of c_{12} already has large positive coefficients and a small negative constant term, suggesting that any positivity statement must occur before full cancellation.

11 Further frontier directions

11.1 A proof program for Jacobi-increment positivity

Conjecture 6.4 is especially promising because it explains a striking combination of facts: the dyadic up-law appears to have positive even free cumulants, its Jacobi parameters appear monotone, yet it is not freely infinitely divisible. A proof program could proceed as follows.

First, express moments as weighted Dyck paths with a down-step from height j weighted by β_j. Second, express free cumulants by a connectedness constraint on those paths, using Lehner's irreducible-lattice-path expansion. Third, mark each occurrence of β_j as $\beta_1 + \Delta_1 + \cdots + \Delta_{j-1}$. The difficulty is to construct a cancellation-free involution or direct class whose weight is the resulting monomial. The monomial counts

$$1, 3, 10, 35, 120, 410, 1400$$

suggest a rapidly growing connected path family and may help identify the correct combinatorial class.

11.2 Determining the full free-divisibility region

The leading-Hankel thresholds in Table 2 appear to stabilize near $q \approx 0.6187$, tantalizingly close to but not exactly the inverse golden ratio. This numerical observation should not be overinterpreted. Leading principal minors are only part of conditional positive definiteness; nonprincipal minors and other test vectors can fail later. A robust investigation should:

(a) compute all principal and selected nonprincipal minors to increasing order;

(b) formulate the conditional free Lévy measure, if one exists, from the free cumulants;

(c) use the reciprocal Cauchy transform to test the Pick/Nevanlinna criterion for free infinite divisibility;

(d) compare the analytic continuation threshold with the Gaussian limit $q \uparrow 1$.

An exact proof of a phase transition would likely need analytic control of the Voiculescu transform, not only moment matrices.

11.3 Free and Boolean transforms of the inverse function

Formula (51) gives finite integral invariants of F^{-1}, but it does not yet exploit the all-orders inverse endpoint expansion. One can insert the repository's Lambert–W expansion of $F^{-1}(u)$ near zero into (50) and then apply the finite cumulant transforms. This should yield asymptotic expansions for large-order free and Boolean cumulants in terms of the same Bell-polynomial coefficient packages already used for the inverse Fabius function.

A particularly concrete target is an expansion

$$r_{2n} = S^{-2n} \exp\left[-\Theta\left(\frac{n}{W(n)}\right)\right] \times \big(\text{explicit Lambert–}W\text{ series}\big), \tag{80}$$

with the correct scale and prefactor to be determined. Equation (80) is a research template, not a claimed asymptotic formula.

11.4 Legendre and Jacobi error transport

Proposition 7.2 is triangular. If a numerical Legendre expansion has coefficient errors δa_{2k}, then

$$\delta m_{2n} = \sum_{k=0}^{n} \frac{2(2n)!}{(2n-2k)!!(2n+2k+1)!!} \delta a_{2k}.$$

Combining this with the Jacobian of the moment-to-free-cumulant map gives certified error bars for r_{2n}. Conversely, the exact rational dyadic cumulants provide nonlinear moment constraints that can detect hidden loss of precision in high-order Legendre or Jacobi computations. This is a practical bridge between the repository's representation volumes and the exact arithmetic developed here.

11.5 Beyond the dyadic base

For $|q| < 1$, symmetry reduces negative q to positive $|q|$, so Theorem 5.2 immediately extends to

$$0 < |q| < 0.523498599534498\ldots.$$

For $|q| > 1$, the infinite series (1) diverges, but finite products can be renormalized by reversing the weights:

$$q^{-N} Y_q^{(N)} = \sum_{j=0}^{N-1} q^{-j} U_{N-j}.$$

This converges to a geometric-uniform law with parameter $1/|q|$. Thus the values $q = \pm 2, \pm 4$ are naturally dual to $q = \pm 1/2, \pm 1/4$ after finite-level reversal. It would be useful to reformulate the repository's $q \leftrightarrow q^{-1}$ identities directly at the level of classical, free, and Boolean cumulants.

11.6 Arithmetic questions

At $q = 1/2$, every moment and cumulant is rational. Their denominators mix Bernoulli denominators, Mersenne-type factors $4^m - 1$, and partition-transform denominators. Possible arithmetic projects include:

(i) sharp p-adic valuations of r_{2n} and b_{2n};

(ii) congruences inherited from von Staudt–Clausen and Kummer congruences for Bernoulli numbers;

(iii) integrality after multiplication by a canonical product of $(4^j - 1)$ and factorial factors;

(iv) denominator stabilization under finite-level Richardson extraction.

The exact positivity through r_{120} supplies a substantial initial data set for such experiments.

11.7 Formal verification

The central non-free-infinite-divisibility proof is well suited to formalization. Its analytic input can be isolated as the standard theorem "free infinite divisibility implies conditional positive definiteness of free cumulants." The remaining proof is finite:

(1) derive the first six even moments from the dyadic cumulant recurrence;

(2) derive $r_2, \ldots, r_{12}$ from noncrossing recurrences;

(3) normalize by variance;

(4) evaluate the rational quadratic form (25).

This would make a compact Lean extension of the repository's current formalized core. The degree-thirty q-family theorem is also formalizable, but requires a certified Sturm or polynomial-root library.

12 Conclusions

The noncommutative cumulant viewpoint reveals a new set of structures hidden inside the familiar sinc product. The up-law has positive low- and medium-order free cumulants but fails the global Hankel positivity required for free infinite divisibility. This failure is exact, rational, and robust across a nontrivial interval of geometric parameters. Boolean cumulants behave more regularly: Jacobi coefficient stripping proves positivity at every order and identifies them with an associated moment sequence.

The strongest conceptual bridge is that the same finite set of moments simultaneously admits four descriptions:

$$\begin{gathered}\text{Bernoulli numbers and Bell polynomials}\\ \updownarrow\\ \text{infinite or finite sinc products and Thue–Morse splines}\\ \updownarrow\\ \text{inverse-Fabius integrals and Legendre coefficients}\\ \updownarrow\\ \text{free and Boolean cumulants and Jacobi stripping.}\end{gathered}$$

The transformations between these descriptions are finite and exact at each order. This makes the new layer compatible with the repository's existing q-binomial acceleration, spectral representations, and Lambert–W asymptotics rather than parallel to them.

The principal open goals are now sharply stated: prove all-order Jacobi-increment positivity, determine the free-divisibility phase diagram of Y_q, and transfer the endpoint Lambert–W expansion through the Boolean reciprocal and free compositional transforms. Each problem has a concrete algebraic or analytic entry point and a reproducible computational base.

13 Reproducibility guide

The package contains `numerical_experiments.py`. It requires Python 3 together with SymPy, mpmath, NumPy, and Matplotlib. From the report directory, run the three stages as separate processes:

```
python3 numerical_experiments.py --task jacobi \
  --output-dir results --jacobi-order 8
python3 numerical_experiments.py --task finite \
  --output-dir results --finite-order 6
python3 numerical_experiments.py --task analysis \
  --output-dir results --figure-dir figures \
  --exact-order 60 --endpoint-order 200 --hankel-max 10
```

Together these three commands perform:

(1) exact rational dyadic cumulants through r_{120};

(2) the determinant and polynomial-vector certificates;

(3) exact symbolic $c_2, \ldots, c_{12}$, the degree-thirty numerator, root isolation, and Sturm counts;

(4) high-precision leading-Hankel thresholds through block size ten;

(5) the Jacobi-increment expansion through r_{16};

(6) finite-level $Q = 4^{-N}$ polynomials through order twelve;

(7) endpoint moments through m_{400}, Boolean ratios, and figures.

All exact claims are checked with rational or integer arithmetic. Decimals attached to exact certificates are printed only after the exact calculation; endpoint and threshold diagnostics instead use the explicit high precision described above. The generated files under `results/` include CSV tables and plain-text certificates; figures are available as both PDF and PNG.

A Explicit normalized q-family formulas

Put $s = q^2$. In addition to Proposition 5.1,

$$D_{10}(s) := (1+s)^2(1+s^2)(1+s+s^2)(1+s+s^2+s^3+s^4),$$
$$c_{10}(s) = \frac{2P_{10}(s)}{385D_{10}(s)}. \tag{81}$$

where

$$P_{10}(s) = 2437711s^{10} + 646150s^9 + 757701s^8 + 780549s^7 + 782870s^6 + 159568s^5 + 136730s^4 + 25179s^3 + 2331s^2 + 10s + 1.$$

Also

$$D_{12}(s) := (1+s)^3(1+s^2)(1-s+s^2)(1+s+s^2)^2 \times (1+s+s^2+s^3+s^4),$$
$$c_{12}(s) = \frac{2P_{12}(s)}{875875D_{12}(s)}. \tag{82}$$

where

$$P_{12}(s) = 239971210481s^{15} + 79887931849s^{14} + 98139745702s^{13} + 106474515277s^{12} + 123041556053s^{11} + 114309825775s^{10} + 52750980414s^9 + 35665790305s^8 + 27408293365s^7 + 10846922526s^6 + 2506486435s^5 + 1249569737s^4 + 82945993s^3 + 5673358s^2 + 3421s - 691.$$

These formulas specialize at $s = 1/4$ to the normalized dyadic cumulants in Section 4 and at $s = 1$ to the connected-pairing numbers 248 and 2830.

B The degree-thirty Sturm polynomial

Define

$$P_{30}(s) = \sum_{j=0}^{30} p_j s^j.$$

The coefficients are as follows.

Table 3: Coefficients of the determinant numerator P_{30}.

j	p_j
0	-1
1	236797
2	154644128
3	-32161726341
4	-309530241402
5	-4962354733999
6	-25423927396444
7	-174414460271621
8	500704487135269

j	p_j
9	-1567422948953118
10	6462787608357749
11	17451160220542367
12	47450105349899789
13	-3112870344293064
14	111335265126207640
15	260832096100041918
16	710585546823441352
17	948703157726602056
18	944899121577446453
19	2149494971059201511
20	3586241741057425853
21	2006573378166739986
22	4423316345984162365
23	2656469118337756915
24	5387333540324201540
25	4570485884957325545
26	6277464174324435942
27	3802457057684963283
28	3106555208559680768
29	1365452791577583829
30	588031288239582935

The three positive roots are isolated in the rational intervals

$$\frac{6261}{1486660088} < s_1 < \frac{7013}{1665220763},$$
$$\frac{17338229}{3019459543} < s_2 < \frac{18557046}{3231717013},$$
$$\frac{385546279}{1406842461} < s_3 < \frac{219810848}{802080713}.$$

Numerically,

$$\begin{aligned} s_1 &= 0.000004211453613733039418535107394599646153802598\ldots, \\ s_2 &= 0.005742163043778858179021510344163665436168765139\ldots, \\ s_3 &= 0.274050783714581103197636083256401808258849748968\ldots. \end{aligned}$$

An exact Sturm sequence gives three roots in $(0, 1)$, no root in $(1/11, s_3^-)$, exactly one in the displayed interval for s_3, and none between its right endpoint and 1.

C Additional finite-level polynomials

For reference, the order-ten and order-twelve free-cumulant polynomials are

$$\begin{aligned} r_{10}^{(N)} &= -\frac{2(Q-1)}{23062937695875}\big(930453829171Q^4 - 628330001950Q^3 \\ &\qquad + 149236761408Q^2 - 14655524050Q + 519975421\big), \\ r_{12}^{(N)} &= \frac{2(Q-1)}{214857210441983484375}\big(32015207756442935833Q^5 \\ &\qquad - 24643616781732391817Q^4 + 7141155293332868434Q^3 \\ &\qquad - 976484255895326266Q^2 + 64569346306285733Q - 1693269622691917\big). \end{aligned}$$

The order-eight through order-twelve Boolean polynomials are

$$
\begin{aligned}
b_8^{(N)} &= \frac{4(Q-1)}{3415820625}\big(42029857Q^3 - 39615618Q^2 + 13585593Q - 1711432\big),\\
b_{10}^{(N)} &= -\frac{4(Q-1)}{115314688479375}\big(3485827553819Q^4 - 3640877184125Q^3\\
&\qquad + 1538558237937Q^2 - 317835338375Q + 27389330744\big),\\
b_{12}^{(N)} &= \frac{4(Q-1)}{2148572104419834843 75}\big(2266378983719264527 9Q^5\\
&\qquad - 24954751286859145796Q^4 + 11665865760722459242Q^3\\
&\qquad - 2968584330536746108Q^2 + 419917784310690479Q - 26668720414063096\big).
\end{aligned}
$$

Every displayed expression has degree equal to its compressed order.

D A compact proof certificate for the dyadic theorem

A verifier need only check the following data:

$$
\begin{aligned}
r_2 &= \frac{1}{9}, & r_4 &= \frac{7}{2025}, & r_6 &= \frac{562}{893025},\\
r_8 &= \frac{55603}{379535625}, & r_{10} &= \frac{148564406}{3294705385125}, & r_{12} &= \frac{3386539245383834}{214857210441983484375}.
\end{aligned}
$$

After multiplying r_{2n} by 9^n, form H in (23). Then

$$
(3,-4,1)H(3,-4,1)^\top = -\frac{108624693817191}{404291747234375} < 0.
$$

This single rational inequality, together with conditional positive definiteness of free cumulants for freely infinitely divisible laws, proves Theorem 4.1.

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